

Introduction to Machine Learning for Environmental Science

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This 2-day workshop introduces core concepts of machine learning including common algorithms, standard workflows and key software libraries. The focus will be on supervised machine learning for regression and classification using a range of methods such as random forests and deep convolutional neural networks. The material will be taught through a mix of theory and practical examples across different environmental sciences such as meteorology and ecology.

Introduction

- Supervised ML
- Model training
- Examples, model evaluation

Tree-based methods

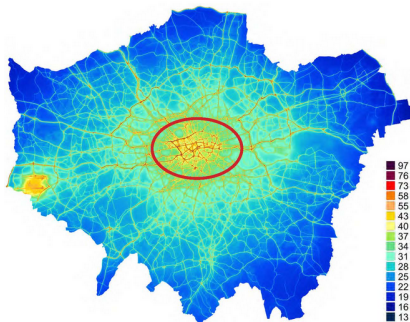
- Decision trees
- Random Forest, XGBoost

Neural Networks

- Multilayer Perceptron
- Convolutional Neural Networks

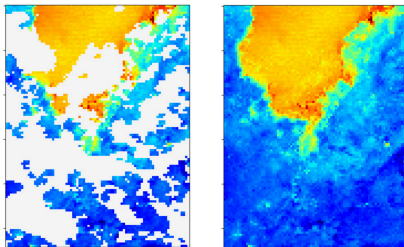
Some examples

Air quality prediction: Predict high-resolution spatial pattern of next day PM10 concentration based on learned relationships between pollution, weather and traffic.



Examples of ML applications

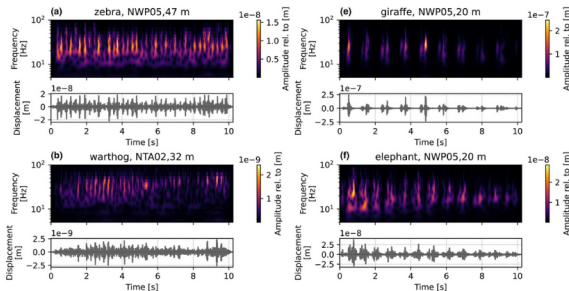
Gap filling: Fill in missing data due to clouds in satellite images based on learned spatial patterns.



<https://doi.org/10.3390/rs12233865>

Examples of ML applications

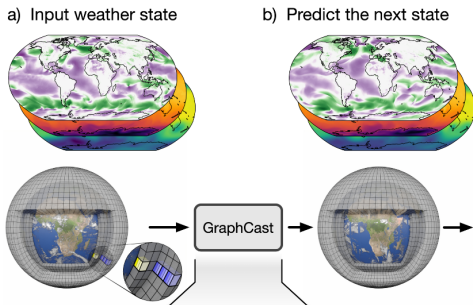
Monitoring wildlife: Learning patterns in seismic signals to detect and monitor animal species.



<https://doi.org/10.1111/2041-210X.70021>

Examples of ML applications

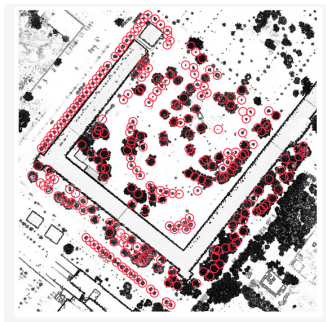
Weather prediction: Learning spatial, temporal and intervariable relationships between weather quantities to make skilful global weather predictions.



<https://doi.org/10.48550/arXiv.2212.12794>

Examples of ML applications

Image segmentation: Detecting and annotating trees on urban maps based on learned image patterns.



<https://doi.org/10.3390/ijgi11040226>

Examples of ML applications

What do these examples have in common?

- High-dimensional data (space, time, multivariable)
- Need to learn complex interactions between variables
- Data-rich but often limited mechanistic understanding
- Clear framing as an input-output relationship
- Focus on skilful prediction of observables

Supervised machine learning: General setup

- We have an observable **target outcome** we wish to predict.
- The prediction can be based on a set of **input features**.
- For example
 - predict future temperature based on current temperature, pressure, wind
 - predict occurrence of wildfire based on measures of heat, humidity, wind
 - predict crop yield based on agricultural practice, soil nutrients, pests
- The goal of supervised ML is to **build a model** that predicts outcomes based on available features.
- A "good" ML model is one that produces **accurate predictions**.

Regression and classification

- The type of target variable determines the type of ML problem.
- **Regression:** Predicting quantitative outcomes.
 - For example: temperature in degrees, crop yield in tonnes, number of sharks, rain amount in mm
- **Classification:** Predicting categorical outcomes.
 - For example: wildfire yes/no, temperature freezing/nonfreezing, precipitation type rain/snow/hail, air quality low/medium/high
- Regression vs classification determines various details of the ML model and analysis.

WIP: Regression and classification

TODO: example outputs from a regression model and a classification model used later in the course to illustrate the distinction

Function approximation theory

- A ML model is a **prediction rule** that translates one or more features $x_1, x_2, \dots$ into an estimate $\hat{y}$ of the target y .
- The prediction rule is calculated by a **mathematical function**

$$\hat{y} = f(x_1, x_2, \dots)$$

- The function $f(\dots)$ usually includes **model parameters** $\theta_1, \theta_2, \dots$ that control how inputs are translated to outputs.
- Defining the feature vector $\mathbf{x} = (x_1, x_2, \dots)$ and parameter vector $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots)$ a trainable ML model is often compactly written as

$$\hat{y} = f(\mathbf{x}; \boldsymbol{\theta})$$

Training data

- Supervised ML works by "learning from examples".
- **Training data** S is a set of input-output examples

$$S = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$$

where

- $\mathbf{x}_i = (x_{i,1}, x_{i,2}, \dots, x_{i,k})$ is the **feature vector** (containing k features) in the i -th example
- y_i is the **target outcome** observed in the i -th example
- n is the **sample size**
- The data set S is used to train our ML model $f(\mathbf{x}; \theta)$ that can predict the outcome y^* for arbitrary feature vector $\mathbf{x}^*$.

Loss function

- A loss function $L(\hat{y}, y)$ is a function that quantifies **how different** the prediction $\hat{y}$ is to the target outcome y .
- Loss functions are typically **negatively oriented**, where lower values indicate better predictions.
- The loss function must be appropriate for the ML task at hand (regression vs classification) and can usually be set when initialising the ML model.
- Otherwise the choice of loss function is **often ambiguous** and difficult to fully justify a particular choice.

Loss function

- **Regression** (where $\hat{y}$ and y are real numbers):
 - Squared-error loss $L(\hat{y}, y) = (\hat{y} - y)^2$
 - Absolute-error loss $L(\hat{y}, y) = |\hat{y} - y|$
- **Binary classification** (where y is 0 or 1 and $\hat{y}$ is the probability for outcome "1"):
 - Cross-entropy loss $L(\hat{y}, y) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$
 - Quadratic loss $L(\hat{y}, y) = (\hat{y} - y)^2$
- **Multivariate classification** (where $y = (0, \dots, 1, \dots, 0)$ is a binary vector indicating which of k classes occurs and $\hat{y}$ is a vector of k class probabilities)
 - Cross-entropy loss $L(\hat{y}, y) = -\sum_{i=1}^k y_i \log \hat{y}_i$
 - Focal loss $L(\hat{y}, y) = -\sum_{i=1}^k y_i (1 - \hat{y}_i)^\gamma \log \hat{y}_i$

Empirical loss minimisation

- We can now **formalise** training of ML model $f(\mathbf{x}; \theta)$ as
 - trying to find model parameters θ such that
 - the model outputs $\hat{y}_1, \dots, \hat{y}_n$ calculated from features $\mathbf{x}_1, \dots, \mathbf{x}_n$
 - achieve minimal average loss when compared to the targets $y_1, \dots, y_n$ in the training data set
 - using an appropriate loss function $L(\hat{y}, y)$.
- **The Empirical Loss Minimisation paradigm for ML:**

$$\hat{\theta} = \underset{\theta}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^n L[f(\mathbf{x}_i; \theta), y_i]$$

- The trained model with parameters $\hat{\theta}$ can then be used to make predictions $\hat{y}^*$ based on new feature vectors $\mathbf{x}^*$

$$\hat{y}^* = f(\mathbf{x}^*; \hat{\theta})$$

In summary ...

To train a ML model by empirical loss minimisation we need:

- a training data set S containing n examples of inputs and corresponding outcomes
- a model function $f(\mathbf{x}; \theta)$ with trainable parameters θ that translates inputs $\mathbf{x}$ into an output $\hat{y}$
- a loss function $L(\hat{y}, y)$ that quantifies how close model output $\hat{y}$ is to the target outcome y
- a search algorithm to find parameters that minimise the loss

The combination of training data set, search algorithm, model function $f(\mathbf{x}; \theta)$ and loss function $L(\hat{y}, y)$ determine the ML approach and **these details should always be reported when documenting your ML model development.**

Code example: Linear regression

- **Features** $x_1, \dots, x_n$: temperatures at different times $t_1, \dots, t_n$
- **Targets** $y_1, \dots, y_n$: temperatures a short time Δ later
 $t_1 + \Delta, \dots, t_n + \Delta$
- **Model**: Linear function with intercept θ_1 and slope θ_2 , such that

$$\hat{y} = f(x; \theta) = \theta_1 + \theta_2 \cdot x$$

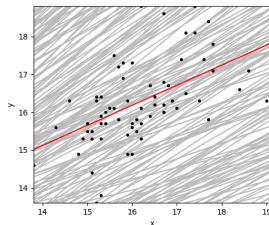
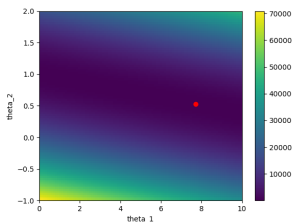
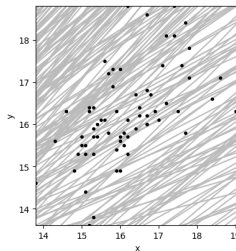
- **Loss**: Squared-error loss

$$L(\hat{y}, y) = (\hat{y} - y)^2$$

- **Search**: Exhaustive search over a grid of parameter values
(don't do this in practice!)

Code example: Linear regression

```
In [129]: np.stack([x,y]).T
Out[129]:
array([[16. , 15.6],
       [15.9, 14.9],
       [15.9, 16.3],
       [15.2, 16.4],
       [13.8, 14.6],
       [16.7, 18.6],
       [15.3, 13.8],
       [16.1, 15.8],
       [15.1, 15.5],
       [16.7, 16.8],
       [14.9, 15.3],
       [15.3, 15.9],
       [15.1, 15.3]
```



```
In [164]: np.stack([x,y,y_hat]).T
Out[164]:
array([[16. , 15.6, 16.18 ],
       [15.9, 14.9, 16.127],
       [15.9, 16.3, 16.127],
       [15.2, 16.4, 15.756],
       [13.8, 14.6, 15.014],
       [16.7, 18.6, 16.551],
       [15.3, 13.8, 15.809],
       [16.1, 15.8, 16.233],
       [15.1, 15.5, 15.703],
       [16.7, 16.8, 16.551],
       [14.9, 15.3, 15.597],
       [15.3, 15.9, 15.809],
       [15.1, 15.3, 15.703],
       [15.1, 14.4, 15.703],
       [15.9, 15.4, 16.127],
       [16. , 15.7, 16.18 ],
       [15.1, 15.3, 15.862]
```

Machine Learning in python with scikit-learn

<https://scikit-learn.org>

Install User Guide API Examples Community More1.7.2 (stable)

scikit-learn

Machine Learning in Python

Getting Started Release Highlights for 1.7

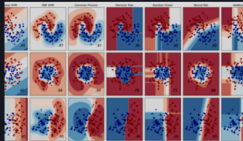
- Simple and efficient tools for predictive data analysis
- Accessible to everybody, and reusable in various contexts
- Built on NumPy, SciPy, and matplotlib
- Open source, commercially usable - BSD license

Classification

Identifying which category an object belongs to.

Applications: Spam detection, image recognition.

Algorithms: [Gradient boosting](#), [nearest neighbors](#), [random forest](#), [logistic regression](#), and [more...](#)



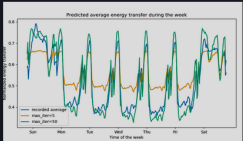
Examples

Regression

Predicting a continuous-valued attribute associated with an object.

Applications: Drug response, stock prices.

Algorithms: [Gradient boosting](#), [nearest neighbors](#), [random forest](#), [ridge](#), and [more...](#)



Examples

Clustering

Automatic grouping of similar objects into sets.

Applications: Customer segmentation, grouping experiment outcomes.

Algorithms: [k-Means](#), [HDBSCAN](#), [hierarchical clustering](#), and [more...](#)



Examples

Linear Regression in Scikit-Learn

```
from sklearn.linear_model import LinearRegression

# data wrangling:
# inputs: X - (n x k) array of n examples, k features
# targets: y - 1d array of length n

# initialise a linear regression model
model = LinearRegression()

# fit the model to data
model.fit(X, y)

# extract parameter estimates from fitted model
theta_hat = np.array([model.intercept_, model.coef_[0] ])

# calculate predictions in training data
y_hat = model.predict(X)

# calculate predictions on new data
x_new = np.linspace(15, 20, 100).reshape(-1,1)
y_new = model.predict(x_new)
```

How good is my model?

- The model outputs $\hat{y}_1, \dots, \hat{y}_n$ must be compared to the targets $y_1, \dots, y_n$ using a suitable error metric.
- For example root mean squared error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

- Our model has $RMSE \approx 0.9$ – Is that "good"?
- Getting $RMSE > 0$ means our model is not perfect, but how bad is it?
- To judge whether the model is useful we should compare it to a suitable benchmark.

Benchmarking

- A benchmark model (or reference model) for performance evaluation is usually one or several of
 - An alternative (competing) model for the same target.
 - Climatology (constant mean):

$$\hat{y}_i^{(clim)} = \frac{1}{n} \sum_{i=1}^n y_i$$

- Persistence (last available observation of the target); here:

$$\hat{y}_i^{(pers)} = y(t_i - \Delta)$$

- Any other simple estimate of the target that could be calculated with reasonable effort given the same inputs as our model.
- In fact, linear regression itself is often a good reference to benchmark more complicated ML model.

Skill scores

- Given a suitable error metric, we calculate
 - S : the error of our model (averaged over a test data set)
 - S_{ref} : the mean error of our chosen reference model
 - S_{perf} : the mean error of a hypothetical perfect model that outputs $\hat{y}_i = y_i$ each time; (usually $S_{perf} = 0$)
- The skill score of our model relative to the benchmark is then defined as

$$Skill = \frac{S_{ref} - S}{S_{ref} - S_{perf}}$$

- $Skill \leq 0$: Our model is not better than the benchmark.
- $Skill \in (0, 1)$: Our model improves over the benchmark.
- $Skill = 1$: Our model is perfect.

Skill scores

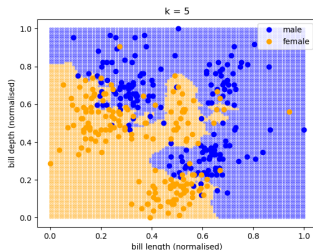
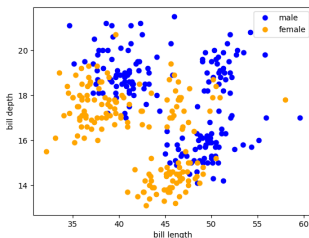
MORE HERE Skill scores of linear regression model

Example: k-nearest-neighbor classification

- given an input $\mathbf{x}^*$ find the k closest features $\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(k)}$ in the training data, and the corresponding outcomes $y^{(1)}, \dots, y^{(k)}$
- "closeness" between the feature vectors $\mathbf{x}^{(i)}$ and $\mathbf{x}^{(j)}$ is defined in terms of the Euclidean distance $\sqrt{\sum_m (x_m^{(i)} - x_m^{(j)})^2}$
- predict $\hat{y}^*$ as the majority vote over $y^{(1)}, \dots, y^{(k)}$
- ties are resolved by picking the outcome class lowest value or alphabetical rank

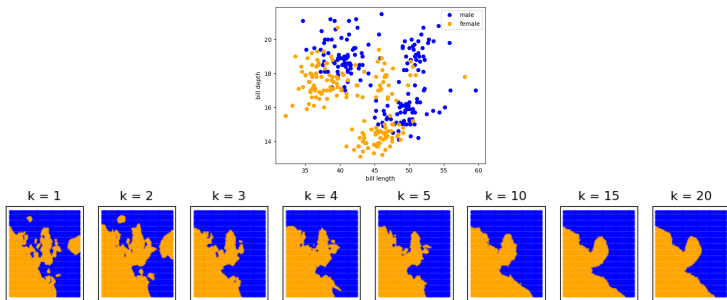
Example: k-nearest-neighbor classification

- For illustration we use data from the "Palmer Penguin" data.
- Features are penguins' bill length (x_1) and bill depth (x_2) and outcome (y) is the penguin's sex (male or female).
- The kNN classifier ($k = 5$) separates the x_1/x_2 plane into regions for "female" and "male", separated by a decision boundary.



Example: k-nearest-neighbor classification

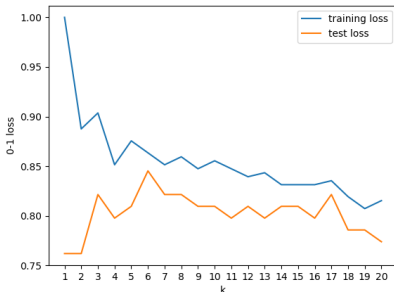
- The parameter k controls the "roughness" of the classifier.
- Larger values of k produce smoother decision boundaries.



- Is k a trainable parameter that can be selected by empirical loss minimisation?

Example: k-nearest-neighbor classification

- Split the data randomly into training and test data set:
 - Only the training data set is used by the kNN classifier.
 - Training and test data are used separately to calculate the 0-1 loss (aka accuracy, aka proportion correct) of the classifier.



- The training accuracy is perfect for $k = 1$ (Why?) and goes down as k increases.
- The test accuracy is best for $k \approx 6$.
- The neighborhood parameter k can be optimised by loss minimisation, but should use a different data set than the one used to define the kNN classifier.

In-sample vs Out-of-sample error

- Our goal is to train a model that generalises well to new data.
- Loss on training data (in-sample error) is often too optimistic because the model "has seen" the data.
- Overfitting: A model achieves very low training loss by "regurgitating" the training data, including any noise and accidental patterns. Overfitted models generalise poorly to new data.
- Loss on previously unseen data (out-of sample error) is what we are really interested in.
- It is good practice to remove a fraction of data (e.g. 20%) during training and use only for model testing.

Further reading

- Linear algebra for least squares regression.
- Alternative loss functions
- Multiple linear regression for multiple inputs.
- Logistic regression for classification.
- Radius-neighborhood classifier and other neighborhood methods.
- Neighborhood methods for missing data imputation.
- Uncertainty estimation: Resampling methods, bootstrapping.

Introduction

- Supervised ML
- Model training
- Examples, model evaluation

Tree-based methods

- Decision trees
- Random Forest, XGBoost

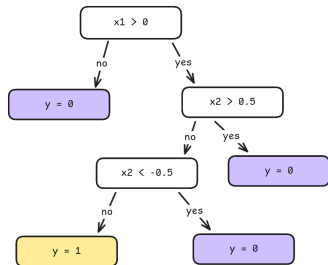
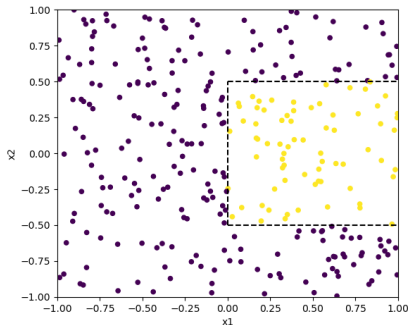
Neural Networks

- Multilayer Perceptron
- Convolutional Neural Networks

Background

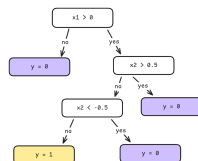
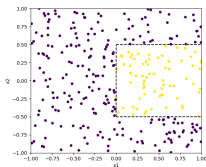
- Tree-based methods are simple and powerful function approximation methods.
- During training, the feature space is split up into rectangular regions.
- The model's prediction within each rectangular region is a constant.

Decision tree example



- Left: Suppose (normalised) precipitation x_1 and temperature x_2 determine survival/death ($y = 1$ or 0) of a plant species by a rectangular "habitable zone" in the $x_1 - x_2$ plane (left).
- Right: The survival/death classification can be represented by a decision tree (right).

Decision tree example



- Classification function for feature vector $\mathbf{x} = (x_1, x_2)$

$$f(\mathbf{x}; \boldsymbol{\theta}) = \begin{cases} 0 & \text{if } x_1 < 0 \\ 0 & \text{if } x_1 > 0, x_2 > 0.5 \\ 0 & \text{if } x_1 > 0, x_2 < -0.5 \\ 1 & \text{otherwise} \end{cases}$$

- If decision thresholds are trainable parameters, a decision tree becomes a ML model for classification.

Decision tree terminology

- **Split:** Partitioning data based on a feature and threshold
- **Node:** a point in the tree where a decision is made
- **Root node:** first split in the tree
- **Internal node:** A node that performs a split
- **Leaf node:** Final node that outputs a prediction
- **Depth:** Number of decision levels from Root node to leaves
- **Impurity:** Measure how mixed the targets are within a split
- **Information gain:** Reduction in impurity after a split

Training a decision tree for binary classification

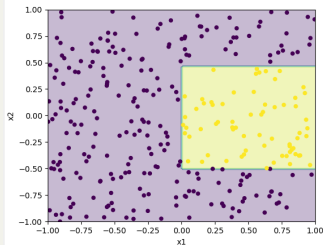
```
from sklearn.tree import DecisionTreeClassifier

# synthetic data
# X.shape = (300, 2) -- x1,x2 as columns
# y.shape = (300,) -- targets

# initialise decision tree with max. depth 3
clf = DecisionTreeClassifier(max_depth=3)

# train decision tree
clf.fit(X, y)

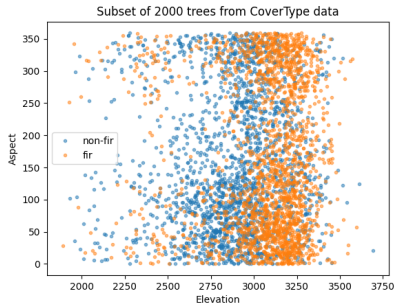
# predict
clf.predict(np.array([0.5,0]).reshape(-1,2))
# returns: array([1])
```



The trained decision tree recovers the correct classification boundary in this simple example.

Code example: Tree cover classification

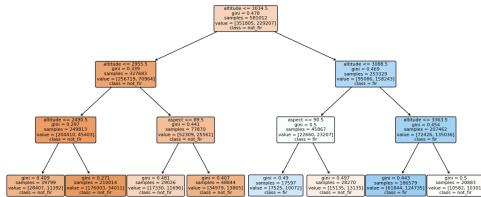
```
import gzip
import numpy as np
path = "covtype.data.gz"
with gzip.open(path, "rt") as f:
    arr = np.genfromtxt(f, delimiter=",")
```



- Cover type data contains data from over 500,000 trees, including topographic, environmental, hydrological, and species information
- Here we wish to predict whether a tree is a pine species, given its height (metres) and aspect (degrees from North)

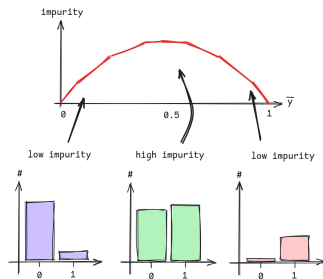
Visualising trained decision trees

```
from sklearn import tree
clf = DecisionTreeClassifier(
    max_depth=3)
clf.fit(X, y)
plt.figure(figsize=(10, 6))
tree.plot_tree(clf,
    feature_names=["altitude",
                   "aspect"],
    class_names=["not_fir",
                 "fir"],
    filled=True, rounded=True,
    fontsize=8)
plt.show()
```



Finding features and thresholds for splitting

- Gini impurity and entropy are measures of impurity, disorder, indeterminism.
- Binary data:
 $\mathbf{y} = (y_1, \dots, y_n) = (0, 0, 1, 0, 1, \dots)$.
- Summarise $\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$.



- Gini impurity: $G(\mathbf{y}) = 1 - \bar{y}^2 - (1 - \bar{y})^2$
- Entropy: $H(\mathbf{y}) = \bar{y} \log \bar{y} + (1 - \bar{y}) \log(1 - \bar{y})$

Maximising Information Gain

- We apply threshold-splitting of the data wrt one feature.
- This separates the data $\mathbf{y}$ into $\mathbf{y}_{left}$ and $\mathbf{y}_{right}$ of sizes n_{left} and n_{right} .
- The impurity of the split data set is calculated by

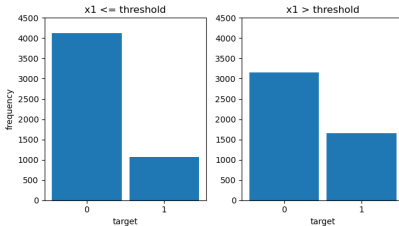
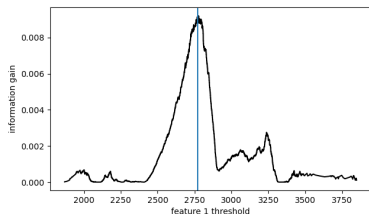
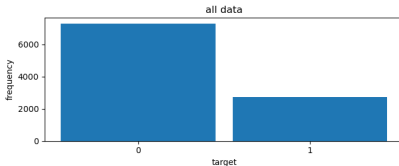
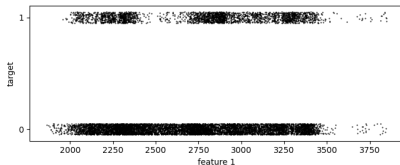
$$G(\mathbf{y}_{left}, \mathbf{y}_{right}) = \frac{n_{left}}{n} G(\mathbf{y}_{left}) + \frac{n_{right}}{n} G(\mathbf{y}_{right})$$

- The impurity reduction (or "information gain" IG) caused by the split is given by

$$IG = G(\mathbf{y}) - G(\mathbf{y}_{left}, \mathbf{y}_{right})$$

- The feature/threshold combination that yields the highest information gain is selected.

Maximising Information Gain



Feature importance

- To interpret a trained ML model it is often interesting to ask "which feature contributes most to the model skill?"
- This is a very difficult question when there are complex interactions between features.
- Some options include
 - Quick and dirty, for decision trees:
`clf.feature_importances_` (mean impurity decrease)
 - More informative and generic:
`sklearn.inspection.permutation_importance`
(performance drop when a feature is randomly shuffled)
 - Widely used, but complicated, measure for reporting: SHAP values (global importance measure)
 - Many more (further reading)

Hyperparameters

- A hyperparameter is a model parameter that is not inferred from the training data by the learning algorithm.
- For decision trees, model parameters are the decision thresholds, and hyperparameters can be
 - minimum/maximum tree depth
 - maximum number of leaf nodes
 - minimum number of samples per split
 - impurity measure



Hyperparameter optimisation

- Cross-validation:
 - randomly split the training data into n_{CV} subsets (usually 5 or 10)
 - use $n_{CV} - 1$ subsets to train the model
 - evaluate the model on the left-out subset
 - repeat for each of the n_{CV} folds
 - average the n_{CV} out-of-sample losses into a single cross-validation loss
- Grid search:
 - Enumerate all n_{comb} combinations of hyperparameters
 - Calculate cross validation loss for each combination
 - Choose the hyperparameter setting with lowest cv loss
 - Fit the model with the optimised hyperparameters
- The computational cost of exhaustive grid search with cross validation is $n_{CV} \times n_{comb}$ larger than training a single model.

Hyperparameter optimisation

```
from sklearn.model_selection import GridSearchCV

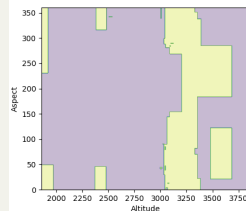
# see all model hyperparameters
clf.get_params()

# initialise grid search
clf_cv = GridSearchCV(
    estimator=clf,
    param_grid={"max_depth": [4,8,16,32],
                "criterion": ["gini", "entropy"]},
    cv=5
)

# run grid search
clf_cv.fit(X, y)

# see best hyperparameters
clf_cv.best_params_

# predict using optimised model
clf_cv.predict(X)
```



Random Forest

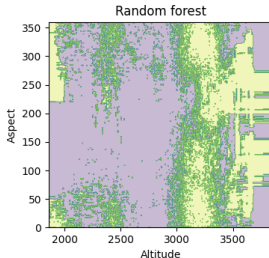
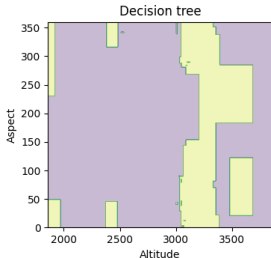
- A RF is an ensemble of indepently trained decision trees.
- Each tree is trained on random subsets of training data and input features.
- Each tree makes a prediction and the forest aggregates them by:
 - Majority vote (Classification)
 - Averaging (Regression)
- Ensembling improves robustness and avoids overfitting compared to single decision tree.

Code example: Random forest

```
from sklearn.ensemble import RandomForestClassifier

# initialise random forest classifier
rfc = RandomForestClassifier(
    n_estimators=100, # number of trees
    max_depth=20,    # maximum tree depth
    random_state=42   # reproducibility
)

# train and predict
model.fit(X, y)
y_pred = model.predict(X)
```



Boosting

- Ensemble methods (such as RF) train a number of **strong** classifiers **independently**.
- **Boosting methods** trains a chain of **weak** classifiers **sequentially**.
- A "weak learner" is simple, fast to train, and has low individual accuracy.
- Core algorithm:
 - train a simple model
 - calculate the model's errors over the training data
 - train the next model to take the same inputs but predict the previous model's errors
 - repeat, then sum the outputs of all models (often with weights)
- Throughout the boosting sequence the model is improved by gradually refining the prediction.

Gradient Boosting

Technical details of gradient boosting:

- **Gradients** of the loss function $g_i = \frac{\partial L(\hat{y}_i, y_i)}{\partial \hat{y}_i}$ approximate prediction errors/residuals.
- Changing $\hat{y}_i$ proportional to $-g_i$ will decrease the loss.
- In **gradient boosting**, each tree predicts the negative gradient $\frac{\partial L(\hat{y}, y)}{\partial \hat{y}}$ of the loss of the previous tree.
- If $f_{i+1}(\mathbf{x}; \theta_{i+1})$ is the output of the $i+1$ -st learner, the $i+1$ st model prediction is

$$\hat{y}_{i+1} = \hat{y}_i + \eta f_{i+1}(\mathbf{x}; \theta_{i+1})$$

where η is the **learning rate**.

XGBoost



- XGBoost is a popular and efficient library for Gradient Boosting models for regression and classification
- Website: `xgboost.ai`
- Very similar naming conventions to scikit-learn:

```
import xgboost as xgb
clf = xgb.XGBClassifier(n_estimators = 100, max_depth=5)
clf.fit(X_train, y_train)
y_hat = clf.predict(X_test)
```

XGBoost for Boosted Decision Trees

```
import xgboost as xgb
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score

# simple XGB model: very weak learners, few boosting rounds
clf_xgb = xgb.XGBClassifier(
    n_estimators = 10,
    max_depth = 1,
    random_state = 42
)

# fit and predict
clf_xgb.fit(X_train, y_train)
y_pred_xgb = clf_xgb.predict(X_test)

# evaluate test data
accuracy_score(y_test, y_pred_xgb)
# accuracy about 0.72

# random forest with same complexity
clf_rf = RandomForestClassifier(
    n_estimators=10, random_state=42, max_depth=1)
clf_rf.fit(X_train, y_train)
y_pred_rf = clf_rf.predict(X_test)
accuracy_score(y_test, y_pred_rf)
# accuracy only about 0.6
```

Evaluation metrics in scikit-learn

- Scikit-learn offers numerous evaluation metrics to compare a set of predictions $\hat{y}_1, \dots, \hat{y}_n$ to a corresponding targets $y_1, \dots, y_n$.
- Some commonly used evaluation metrics/methods are
 - Accuracy
 - Confusion matrix
 - Classification report
 - ROC/AUC analysis
- See scikit-learn.org/stable/modules/model_evaluation.html for a full list and details.

Introduction

- Supervised ML
- Model training
- Examples, model evaluation

Tree-based methods

- Decision trees
- Random Forest, XGBoost

Neural Networks

- Multilayer Perceptron
- Convolutional Neural Networks

Artificial neuron (Perceptron)

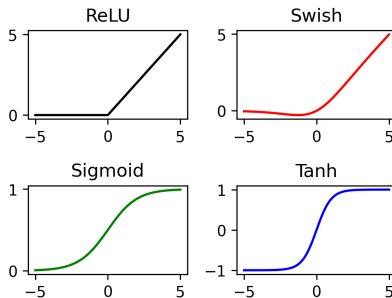
- An artificial neuron (perceptron) is a mathematical function

$$\hat{y} = f(\mathbf{x}; \mathbf{w}, b) = \varphi(w_1 \cdot x_1 + \dots + w_k \cdot x_k + b)$$

- where
 - $\mathbf{x} = (x_1, \dots, x_k)$ is the k -dimensional input vector
 - y is the 1-dimensional output of the neuron
 - $\mathbf{w} = (w_1, \dots, w_k)$ is the k -dimensional vector of weights
 - b is the bias
 - $\varphi(\cdot)$ is the activation function
- A neuron takes inputs $\mathbf{x} = (x_1, \dots, x_k)$, calculates a **linear combination**, and applies a **non-linear activation function**.

Activation functions

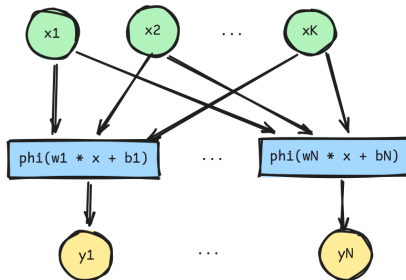
- Typical activation functions:
 - ReLU (Rectifier): $\varphi(x) = \max(0, x)$
 - Sigmoid: $\varphi(x) = 1/(1 + e^{-x})$
 - Swish: $\varphi(x) = x/(1 + e^{-x})$
 - Tanh: $\varphi(x) = (e^x - e^{-x})/(e^x + e^{-x})$



Single-layer perceptron

- In a SLP, **several perceptrons** (each with different weights and biases) calculating several outputs in parallel
- produce a **vector of outputs** $y_1, \dots, y_N$ where

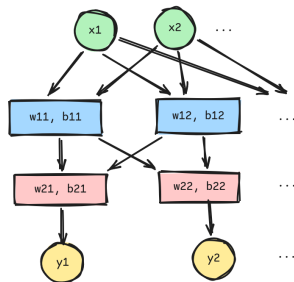
$$y_i = \varphi(w_{i,1}x_1 + \dots + w_{i,k}x_k + b_i) = \varphi(\mathbf{w}_i \cdot \mathbf{x} + b_i)$$



Multilayer perceptron

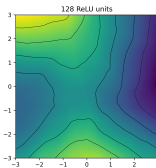
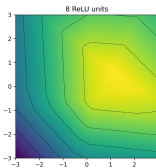
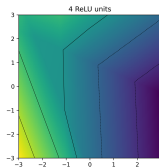
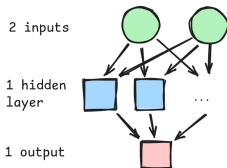
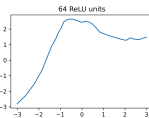
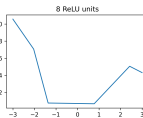
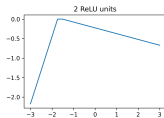
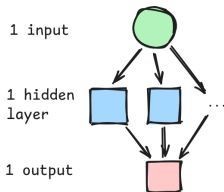
- In an MLP, **perceptrons are arranged in multiple layers**
- Each layer computes a **vector of activations** that serves as input to the next layer
- For layer ℓ , the vector of outputs $\mathbf{y}^{(\ell)} = (y_1^{(\ell)}, \dots, y_{n_\ell}^{(\ell)})$ is calculated from the previous one by

$$y_i^{(\ell)} = \varphi \left(\mathbf{w}_i^{(\ell)} \cdot \mathbf{y}^{(\ell-1)} + b_i^{(\ell)} \right)$$



Examples

Randomly initialised ReLU networks with one hidden layer:



More nodes (units) produce more "flexible" curves.

Terminology

- **Weight:** trainable parameter scaling an input
- **Bias:** trainable offset added to the weighted sum
- **Pre-activation:** linear combination $z = w \cdot x + b$
- **Activation function:** nonlinear function applied to pre-activation
- **Input layer:** First layer that accepts the input features
- **Output layer:** Final layer that produces the model output(s)
- **Hidden layer(s):** layer(s) between input and output layers
- **Parameters:** all trainable weights and biases of the network
- **Forward pass:** computation of outputs from inputs by the network

PyTorch, TensorFlow

- PyTorch and Tensorflow are "low-level" frameworks for implementing neural networks.
- Automatic differentiation and backpropagation.
- Fast training on multicore CPUs and GPUs "out of the box".
- PyTorch is used in research, TF more widely used in production.
- Both can be used as a high-performance alternative to numpy.
- See pytorch.org and tensorflow.org.



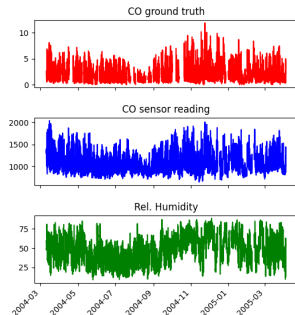
Keras

- High-level API for building NNs, based on TensorFlow.
- Simple syntax and functions to compose "standard NN layers" into complex deep NN architectures.
- Ideal for learning and prototyping, and commonly used in research and applications.
- See keras.io.



Worked example: Air quality prediction

- **UCI Air quality data:** Hourly urban gas concentration readings
- **Ground truth (GT) concentrations:** CO (column 3), Non-methanic hydrocarbons NMHC (5), Benzene C₆H₆ (6), NO_x (8), NO₂ (10)
- **Sensor readings** PT08.S1...5 - CO (4), NMHC (7), NO_x (9), NO₂ (11), O₃ (12)
- **Meteorology:** Temp. (13), Rel. Humidity (14), Abs. Humidity (15)
- **Goal:** Train a NN that predicts ground truth of one pollutant from all sensor readings and meteorological variables.



Data Source: <https://archive.ics.uci.edu/dataset/360/air+quality>

Worked example: Air quality prediction

Workflow overview:

- Load raw data, parse data types
- Define target and feature columns
- Remove rows with missing data
- Construct target and feature variables
- Train-test split
- Feature scaling
- Initialise model architecture and optimiser
- Train model, while tracking validation loss
- Evaluate model on test data
- Benchmark against climatology and linear regression

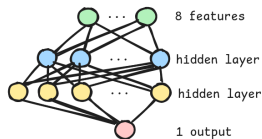
Worked example: Air quality prediction

```
# define MLP model
model = keras.Sequential([
    Input(shape=(X_train.shape[1],)),
    Dense(32, activation="relu"),
    Dense(16, activation="relu"),
    Dense(1)
])

# compile
model.compile(optimizer="adam", loss="mse")

# model summary
model.summary()
# Model: "sequential_11"

# Layer (type)           Output Shape           Param #
# -----
# dense_31 (Dense)       (None, 32)             288
# dense_32 (Dense)       (None, 16)             528
# dense_33 (Dense)       (None, 1)              17
# -----
# Total params: 833 (3.25 KB)
# Trainable params: 833 (3.25 KB)
# Non-trainable params: 0 (0.00 Byte)
```



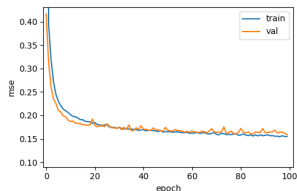
Worked example: Air quality prediction

Train model:

```
history = model.fit(X_train, y_train,  
                    epochs=200, batch_size=32,  
                    validation_split=0.2)
```

Plot learning curve:

```
plt.plot(history.history["loss"],  
         label="train")  
plt.plot(history.history["val_loss"],  
         label="val")  
plt.xlabel("epoch")  
plt.ylabel("mse")  
plt.legend()
```



Worked example: Air quality prediction

Evaluation, benchmarking:

```
# model predictions and test MSE
yhat_test = model(X_test)
mse = np.mean(np.square(y_test - yhat_test))
# around 0.15 #####

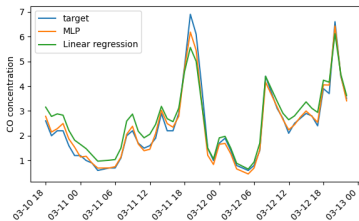
# climatology benchmark
yhat_clim = np.mean(y_train) * np.ones_like(y_test)
mse_clim = np.mean(np.square(y_test - yhat_clim))
# around 1.95 #####

# multiple linear regression benchmark
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr.fit(X_train, y_train)
yhat_lr = lr.predict(X_test)
mse_lr = np.mean(np.square(yhat_lr - y_test))
# around 0.2 #####
```

- Much better than climatology
- 25% skill compared to linear regression on the same features.

Worked example: Air quality prediction

Example model output



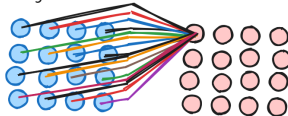
Next steps:

- Retrain the NN with a different architecture (hidden layers, nodes per layer, activation function).
- Retrain the NN with different subsets of features.
- Document changes in performance and interpret the results.

Convolution

- Perceptrons are **densely** **connected** layers because every node in a layer receives outputs from all nodes in the previous layer.
- When inputs and outputs are high-dimensional, the number of trainable parameters in a dense layer becomes very large.
- Convolution is a sparsely connected layer with where outputs are only connected to **spatially close inputs** and weights are **shared across the domain**.

Densely connected with individual weights

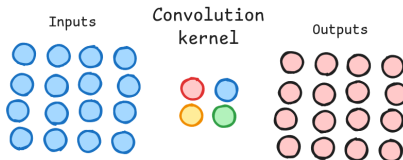


Locally connected with shared weights



Convolution

- The local, shared combination weights are stored in a **convolution kernel**
- Usually kernels are 3×3 or 5×5 , we're using the less common 2×2 for illustration here.
- Boundary conditions need to be considered later.



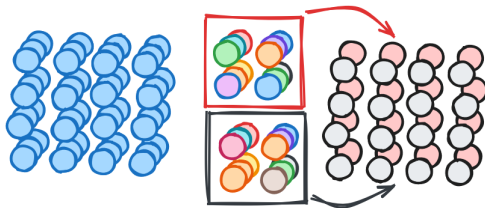
Convolution

- The input data usually have **multiple channels** (variables, color channels, pressure levels)
- The convolution kernel must have the same number of channels as the input data.



Convolution

- By defining multiple convolution kernels, we produce **multiple output channels** from the same input channels.



- This is the **typical situation for 2d convolution** in a convolutional neural network.
- The **trainable parameters** in a 2d convolution are stored in a 4d data structure with dimensions (width, height, input channels, output channels).

Convolution

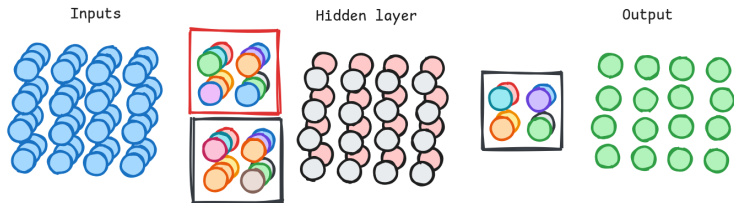
Convolution mathematically:

- inputs $x[i, j, c]$ indexed by space (i, j) and input channel $c = 1, \dots, C$
- outputs $y[i, j, d]$ indexed by space (i, j) and output channel $d = 1, \dots, D$
- kernel $k[m, n, c, d]$ of width and height $m = -w, \dots, +w$, C input channels, D output channels

$$y[i, j, d] = \sum_{m=-w}^w \sum_{n=-w}^w \sum_{c=1}^C k[m, n, c, d] \cdot x[i + m, j + n, c]$$

Convolution

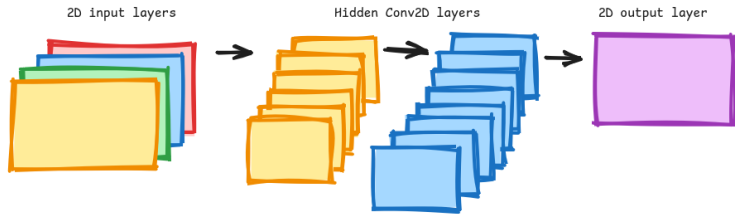
- 2D-convolutions can be chained into layers of a **Deep Convolutional Neural Network** (Deep CNN).



Typical setup

- A common situation in ML with spatial data:
 - several input variables, each a 2d spatial field (input channels)
 - the target is a single spatial field of the same size (output channel)
- A typical CNN for such data has:
 - 2-6 hidden convolution layers
 - kernel size 3 or 5
 - 16-256 channels per hidden layer
 - 1 channel at output layer (to match the target)
- This will produce models with 10^4 - 10^6 parameters, suitable for training locally on moderately large training data sets
- These hyperparameters can be explored and optimised as part of the ML model development.

CNN example applications

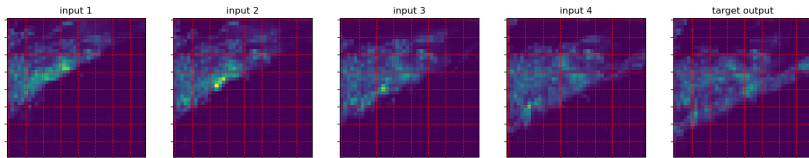


- Spatial classification of vegetation types
- Hazard maps
- Image segmentation
- Next-time step prediction

Worked example: Radar nowcasting

Setup:

- We want to train CNN for next frame image prediction.
- Training data: 1000 randomly sampled radar image sequences
- Input: tensor of shape $(40, 40, 4)$ – 4 consecutive rain radar frames of size 40×40 (resolution 5km), 15 minutes apart.
- Target: tensor of shape $(40, 40, 1)$ – the next frame 15min ahead



Worked example: Radar Nowcasting

```
# input and output data
movies = np.load('radar_movies.npy') # shape (1000,40,40,20)
x = movies[:, :, :, 0:4] # inputs: 4 frames
y = movies[:, :, :, 4:5] # output: 5th frame
```

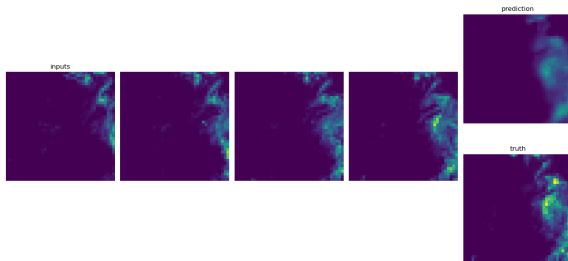
```
# 3 hidden (3x3) conv2d layers each with 32 units
model = Sequential([
    Input(shape=(40,40,4)),
    Conv2D(32, (3,3), activation='relu', padding='same'),
    Conv2D(32, (3,3), activation='relu', padding='same'),
    Conv2D(32, (3,3), activation='relu', padding='same'),
    Conv2D(1, (3,3), activation='relu', padding='same')
])
```

```
model.summary()
# Trainable params: 19,969 (78.00 KB)
model.output_shape
# (None, 40, 40, 1)

# compile and train the model
model.compile(optimizer='adam', loss='mse')
model.fit(x_train, y_train)
```

Worked example: Radar Nowcasting

Visualising the prediction:



Worked example: Radar Nowcasting

Benchmarking the model:

```
# calculate model output on test data
y_test_pred = model.predict(x_test)

# calculate test MSE
np.mean((y_test_pred - y_test)**2)
# 532.405

# calculate the "persistence" MSE as benchmark
np.mean((x_test[:, :, :, 3:] - y_test) ** 2)
# 945.156
```


Training Neural Networks

Gradient descent algorithm:

- ① Initialise model weights randomly.
- ② Repeat until convergence:
 - Calculate **gradient** of the loss function with respect to model weights:

$$\frac{\partial L}{\partial \mathbf{w}}(\mathbf{w}) = \begin{pmatrix} \frac{\partial L(\mathbf{w})}{\partial w_1} \\ \vdots \\ \frac{\partial L(\mathbf{w})}{\partial w_k} \end{pmatrix}$$

- **Adjust weights** in the opposite direction of the gradient:

$$\mathbf{w}^{t+1} \leftarrow \mathbf{w}^t - \gamma \frac{\partial L}{\partial \mathbf{w}}(\mathbf{w}^t)$$

where γ is called the **learning rate**

- ③ Return weights.

Stochastic Gradient Descent (SGD)

- In Gradient Descent, the loss and its gradient are calculated on the whole training data set.
- In SGD, loss and gradient are calculated on random subsets (**batches**) of the whole training data set.
- Random subsetting reduces computational cost of gradient calculation and improves convergence of the optimiser.
- The time taken to loop over the entire training data set once is called an **epoch**.
- A variant of SGD called the **Adam optimiser** is commonly used.

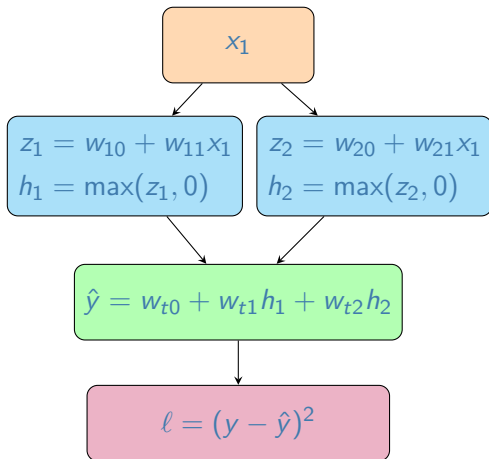
```
# optimiser, loss are specified during initialisation
model.compile(optimizer='adam', loss='mse')

# epochs and batch size are specified during training
model.fit(X, y, epochs=100, batch_size=64)
```

Backpropagation

- "Backprop" is an efficient algorithm to calculate the gradient of the loss with respect model weights
- Propagates loss backward from output to input layers using the chain rule of differentiation
- Efficiency by reusing intermediate computations
- More efficient than, e.g., finite differencing which requires separate loss evaluations for each weight
- Backpropagation exploits automatic differentiation in modern ML libraries such as PyTorch and TensorFlow

Backpropagation example



- gradient of the loss ℓ wrt w_{11} is

$$\frac{\partial \ell}{\partial w_{11}} = \frac{\partial \ell}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial h_1} \times \frac{\partial h_1}{\partial z_1} \times \frac{\partial z_1}{\partial w_{11}}$$

- need computational graph and derivatives
- PyTorch/Tensorflow do this automatically when defining a network

Important hyperparameters

Important hyperparameters to explore next:

```
Dense(units, activation, use_bias,  
      kernel_initializer, bias_initializer, kernel_regularizer)  
  
model.compile(optimizer, loss, metrics)  
  
model.fit(x, y,  
         batch_size, epochs,  
         initial_epoch, steps_per_epoch,  
         validation_split, validation_data, validation_freq,  
         callbacks,  
         workers, use_multiprocessing  
)
```

Early stopping

- When validation loss gets worse we are overfitting.
- We should stop model training early and use the model with lowest validation loss found so far.

```
from tensorflow.keras.callbacks import EarlyStopping  
  
early_stopping = EarlyStopping(monitor="val_loss",  
                                patience=2, restore_best_weights=True)  
  
model.fit(X_train, y_train,  
          validation_split=0.2, epochs=500, batch_size=32,  
          callbacks=[early_stop])
```



Regularisation

Weight-level methods to avoid overfitting:

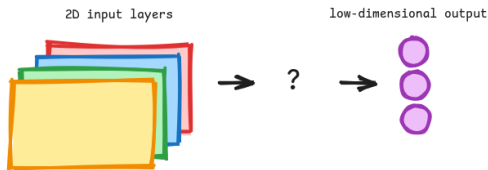
- **Weight regularisation:** Add a term to the loss function that is proportional to the magnitude of the weights. Encouraging the optimiser to keep the weights small can avoid overfitting.
- **Dropout:** During training a random subset of layer outputs are set to zero. The network learns "stable" representations of the data, and generalises better.

```
from tensorflow.keras.layers import Dropout
from tensorflow.keras import regularizers

model = Sequential([
    Input(input_shape=(X_train.shape[1],)),
    Dense(64, activation="relu", kernel_regularizer=regularizers.l2(1e-4)),
    Dropout(0.3),
    Dense(32, activation="relu", kernel_regularizer=regularizers.l2(1e-4)),
    Dropout(0.3),
    Dense(1) ])
```

CNN for image classification

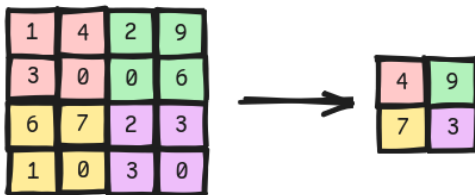
- Goal: Learn to assign one of k labels to an input image
- Application: Classify the animal shown in a photograph
- Output: vector of length k containing class probabilities
- Need to be able to reduce dimension of input to produce the output



Pooling Layer

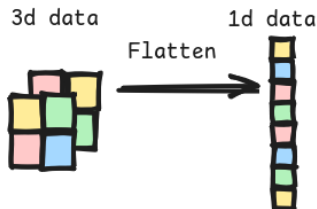
- A pooling layer reduces spatial resolution by calculating windowed summaries
- Max-pooling is most common, but average-pooling, variance-pooling and others exist
- Pooling layers (usually) have no trainable parameters.

2x2 Max-Pooling



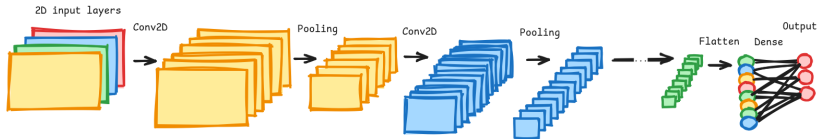
Flatten Layer

- A flatten layer reshapes the input by collapsing all spatial and channel data into a 1d vector
- Flatten has no trainable parameters.



Encoder architecture

- An encoder architecture is a chain of convolution and pooling layers, usually ending in a densely connected MLP, to transform high-dimensional inputs to low dimensional outputs.



Training a multi-class classifier

Softmax output

- We want our model to output a vector of probabilities that sum to one.
- This is achieved by the **softmax activation function** that takes arbitrary inputs $x_1, \dots, x_k$ and outputs a vector with i -th element

$$\text{softmax}(x_1, \dots, x_k)_i = \frac{e^{x_i}}{\sum_{j=1}^k e^{x_j}}$$

Training a multi-class classifier

One-hot-encoding

- To train a multi-class classifier, we one-hot-encode our target labels.
- If a target takes 1 out of k possible classes, the one-hot-encoding is a vector of length k , where the element in the target class position is 1 and all other elements are 0
- For example, when there are 5 possible classes, a target with class label 4 is one-hot-encoded by the vector

$$\mathbf{y} = (0, 0, 0, 1, 0)$$

Training a multi-class classifier

Categorical cross entropy loss

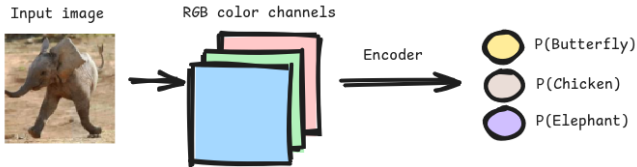
- Cross entropy is a loss function to train classifiers that output a vector of probabilities $\hat{\mathbf{y}} = (\hat{y}_1, \dots, \hat{y}_k)$ of length k and the corresponding target $\mathbf{y} = (y_1, \dots, y_k)$ is one-hot-encoded.
- Cross entropy loss is given by

$$L(\hat{\mathbf{y}}, \mathbf{y}) = - \sum_{i=1}^k y_i \log(\hat{y}_i)$$

- In words: The negative logarithm of the probability assigned to the correct class.

Example: Animal image classification

- We want to train an encoder-based image classifier that takes as input an RGB image of dimension 100x100 pixels and outputs a vector of 3 probabilities for class labels "butterfly", "chicken", and "elephant"



Example: Animal image classification

Loading, scaling, one-hot-encoding:

```
# load the stored data
images = np.load('data/animal-images.npy', allow_pickle=True) # shape
          (1500,100,100,3)
labels = np.load('data/animal-labels.npy', allow_pickle=True) # shape (1500,)

# min-max scaling
X = images.astype("float32") / 255.0

# one-hot encoding of class labels
le = LabelEncoder()
y_int = le.fit_transform(labels) # vector of 0/1/2
ohe = OneHotEncoder(sparse_output=False)
y = ohe.fit_transform(y_int.reshape(-1,1)) # matrix with 3 columns

# array([[0., 1., 0.],
#        [0., 1., 0.],
#        [0., 1., 0.],
#        ...,
#        [1., 0., 0.],
#        [1., 0., 0.],
#        [1., 0., 0.]])
```


Example: Animal image classification

Define model, optimiser, loss function

```
# define encoder architecture
model = Sequential([
    Input(shape=X_train.shape[1:]),
    Conv2D(64, (3, 3), activation="relu"),
    MaxPooling2D((2, 2)),
    Conv2D(32, (3, 3), activation="relu"),
    MaxPooling2D((2, 2)),
    Conv2D(16, (3, 3), activation="relu"),
    MaxPooling2D((2, 2)),
    Flatten(),
    Dense(128, activation="relu"),
    Dense(y.shape[1], activation="softmax")
])

# define optimiser and loss function
model.compile(
    optimizer="adam",
    loss="categorical_crossentropy",
    metrics=["accuracy"]
)
```

Note: The combination of softmax activation at the output layer and categorical cross entropy loss ensures that the model is trained to output a vector of class probabilities

Example: Animal image classification

Train with early stopping

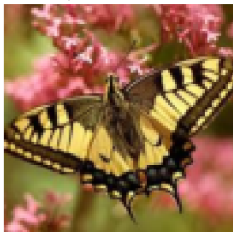
```
early_stop = EarlyStopping(  
    monitor="val_loss", patience=5, restore_best_weights=True)  
  
model.fit(X_train, y_train, epochs=50, batch_size=32,  
    validation_split=0.2, callbacks=[early_stop])  
  
# Epoch 1/50  
# loss: 1.05 - accuracy: 0.39 - val_loss: 0.88 - val_accuracy: 0.59  
# Epoch 2/50  
# loss: 0.82 - accuracy: 0.60 - val_loss: 0.70 - val_accuracy: 0.72  
# .....  
# Epoch 7/50  
# loss: 0.45 - accuracy: 0.81 - val_loss: 0.47 - val_accuracy: 0.74  
# .....  
# Epoch 12/50  
# loss: 0.22 - accuracy: 0.91 - val_loss: 0.62 - val_accuracy: 0.80  
# Epoch 13/50  
# loss: 0.17 - accuracy: 0.94 - val_loss: 0.48 - val_accuracy: 0.83
```

- Optimiser stops early after 13 epochs.

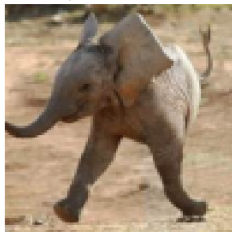
Example: Animal image classification

Some good results:

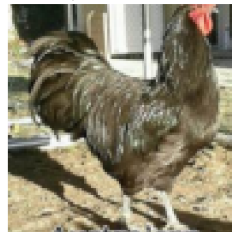
butterfly: 0.987 <-- correct
chicken: 0.013
elephant: 0.000



butterfly: 0.001
chicken: 0.003
elephant: 0.995 <-- correct



butterfly: 0.011
chicken: 0.798 <-- correct
elephant: 0.191



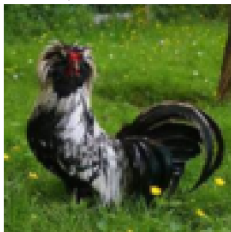
Example: Animal image classification

Some not so good results:

butterfly: 0.134 <-- correct
chicken: 0.758
elephant: 0.109



butterfly: 0.767
chicken: 0.196 <-- correct
elephant: 0.037



butterfly: 0.005
chicken: 0.009 <-- correct
elephant: 0.986

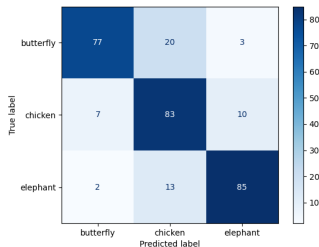


Example: Animal image classification

Confusion matrix

```
# calculate most likely class
y_pred_prob = model.predict(X_test)
y_pred = np.argmax(y_pred_prob, axis=1)
y_true = np.argmax(y_test, axis=1)

# confusion matrix
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=le.classes_
)
disp.plot(cmap="Blues", values_format="%d")
```



Further Reading

- Sensitivity analysis, interpretability
- Fine-tuning for new outputs/objectives
- Speeding up NNs: Quantisation, layer-wise optimisation
- MORE HERE